1_Power

BUCK 12V – 4V5

2A time-delay JFC1032-1200TS

J2 PJ-002BH

F1 B540C-13-F

U1 TPS563203DRLR

L3 4.7uH FXL0530-4R7-M

R1 100K/0603/1%

R45 196k ±1%

R46 30K/0603/1%

R73 5k6R 1%

D33 LED_Red

C1, C2 10uF 25V X5R ±10%

C52, C53 22uF 25V X5R ±20%

C54 100nF 50V X7R ±10%

C55 18pF 50V C0G ±2%

C56 100nF 50V X7R ±10%

C3 100nF 50V X7R ±10%

BUCK 12V – 6V

U2 TPS563203DRLR

L5 4.7uH FXL0530-4R7-M

R4 100K/0603/1%

R70 270K/0603/1%

R74 30K/0603/1%

R75 1K/0603/1%

D34 LED_Red

C8, C9 10uF 25V X5R ±10%

C56, C57 22uF 25V X5R ±20%

C58 18pF 50V C0G ±2%

C59 100nF 50V X7R ±10%

C4 100nF 50V X7R ±10%

BUCK 12V – 3V3

U3 TPS563203DRLR

L2 4.7uH FXL0530-4R7-M

R7 100K/0603/1%

R76 30K/0603/1%

R77 330/0603/1%

D35 LED_Red

C1, C16 10uF 25V X5R ±10%

C60, C61 22uF 25V X5R ±20%

R61 135k ±0.1%

C42 18pF 50V C0G ±2%

C63 100nF 50V X7R ±10%

C5 100nF 50V X7R ±10%

INPUT PROTECTION — REGULATED 12 V ONLY
13.8 V clamp; <2 A current limit; ~10 ms soft start
Validate startup, hot plug and simultaneous loads.

C65 1uF 50V X7R ±10%

C66 100nF 50V X7R ±10%

SMBJ13A

U12 TPS25947ZARPWR

R78 680k ±1%

R79 100k ±1%

R80 39.2k ±1%

R81 1.65k ±1%

C64 3.3nF 50V X7R ±10%

C67 4.7uF 50V X7R ±10%

PWR_FLAG

3_PERIPHERAL

U4 TCA9534APWR

R11 4K7/0603/1%

R13, R14 10K/0603/1%

C22 10nF 50V ±10%

R10 2K2/0603/1%

TP1 TestPoint

C47 100nF 50V X7R ±10%

7-bit I²C address: 1 A5 A4 A3 A2 A1 A0 (R/W sent separately)

Open = 0 / Closed = 1

JP2 SolderJumper_2_Open A0

JP3 SolderJumper_2_Open A1

JP4 SolderJumper_2_Open A2

JP5 SolderJumper_2_Open A3

JP6 SolderJumper_2_Open A4

JP7 SolderJumper_2_Open A5

TP4 TestPoint

TP5 TestPoint

TP6 TestPoint

TP7 TestPoint

TP8 TestPoint

TP9 TestPoint

R29 10K/0603/1%

R30 10K/0603/1%

R31 10K/0603/1%

R32 10K/0603/1%

R33 10K/0603/1%

R34 10K/0603/1%

U7 PCA9685PW,118

C30 100nF 50V X7R ±10%

C29 10uF 25V X5R ±10%

R23 10K/0603/1%

R24 330/0603/1%

D15 LED_Red

R20 4K7/0603/1%

R15 4K7/0603/1%

J9, J12, J17, J3 JST SH 4-pin 1.00mm

J18 JST EH 3-pin 2.50mm

D6, D7, D8 LED_Red

R24, R25, R26 1K/0603/1%

MASTER / EXTERNAL MUX: GND, SDA, SCL only
Each board powered from its own 12 V input.
Exciting 4-pin connector local 3V3 OUTPUT for peripherals.
Multiple boards: TCA9548A at 0x71, one channel active.
Disable PCA9685 ALLCALL after initialization.

2_Pump / Valve Drivers

D13, D14, D17, D19 LED_Red

D11, D12, D16, D18 SS34

U8, U11, U9, U10

Q3, Q4, Q8, Q10 AO3400A

R35, R40, R54, R59 100R/0603/1%

R37, R38, R56, R58 10K/0603/1%

C32, C34, C43, C45 22uF 50V ±20%

C31, C33, C42, C44 100nF 50V X7R ±10%

JST XH 2-pin 2.50mm

ZR370-02PM / 4.5V / 2.5 LPM / ~500 mA

Rated voltage	DC 6.0V
Use voltage	DC 5.0V~6.0V
Load current	< 500mA

額定電壓 (Rated voltage) : DC 6.0V

使用電壓 (Operating voltage) : DC 5.0V~6.0V

4_Switch Inputs

SW1, SW2, SW3, SW4

R16, R17, R18, R19 10K/0603/1%

C23, C24, C25, C26 100nF 50V X7R ±10%

5_Pressure Sensor

U5 MPRLS0025PA00001A

C28 10uF 25V X5R ±10%

C27 1nF 50V X7R ±5%

C46 100nF 50V X7R ±10%

R22 10K/0603/1%

TP3 TestPoint_RES

TP2 TestPoint_EOC

Sheets: /	
File: 1.kicad_sch	
Title:	
Size: A1	Date:
KiCad E.D.A. 10.0.6	Rev:
	Id: 1/1

H1 MountingHole
H2 MountingHole
H3 MountingHole
H4 MountingHole